# STAT 140: Design of Experiments
# Section 8: Crossover and Block Randomized Experiments

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## 1 Part I: Crossover Randomized Experiments

### 1.1 Motivation

In the completely randomized experiments of earlier modules, each participant receives exactly one treatment. Inference is based on comparisons *between* different people assigned to different treatments. This introduces between-person variability as a source of noise: the comparison of person $i$ (who received treatment) with person $k$ (who received control) is confounded by any pre-existing differences between $i$ and $k$.

A **matched-pair design** reduces this noise by pairing similar individuals and randomizing treatment within pairs. A **crossover design** goes one step further: every participant receives *both* treatment and control, in different time periods. Each participant therefore serves as their own control, completely eliminating between-person variability from the comparison.

**Why crossover is especially valuable for pain and subjective outcomes.** Pain and other subjective experiences (fatigue, nausea, mood) have no objective external scale. What counts as "moderate pain" for one person may be "severe pain" for another. By having each person experience both conditions and computing within-person differences, crossover designs side-step this inter-individual scaling problem entirely.

### 1.2 Setup and Notation

Consider a population of $N$ participants. Denote the two periods by $j \in \{1, 2\}$. Let $W_{i,j}$ be the treatment indicator for participant $i$ at period $j$: $W_{i,j} = 1$ if participant $i$ receives the active treatment at period $j$, and $W_{i,j} = 0$ if they receive control.

The defining constraint of a crossover design is:

$$W_{i,j=2} = 1 - W_{i,j=1}.$$

This means a participant who receives treatment first (sequence $(1, 0)$) must receive control second, and vice versa (sequence $(0, 1)$). No one receives the same condition in both periods.

### 1.3 The Assignment Mechanism

Let $\mathbf{W}$ be the $N \times 2$ assignment matrix whose $i$-th row is $W_i = (W_{i,j=1}, W_{i,j=2})$.

**Bernoulli assignment:** Each participant independently flips a fair coin to determine their sequence:

$$W_{i,j=1} \sim \text{Bernoulli}\left(\tfrac{1}{2}\right), \qquad W_{i,j=2} = 1 - W_{i,j=1}.$$

The assignment mechanism is:

$$P(\mathbf{W} \mid \mathbf{X}, \mathbf{Y}(0), \mathbf{Y}(1)) = \begin{cases} \frac{1}{2^N} & \text{if } \mathbf{W} \in \mathbb{W}^+, \\ 0 & \text{otherwise,} \end{cases}$$

where $\mathbb{W}^+$ is the set of all $2^N$ valid assignments satisfying the crossover constraint. There are $2^N$ equally probable assignments.

**Completely randomized assignment:** Fix in advance the number of participants $N_{T,j=1}$ who receive treatment first. Draw uniformly from all $\binom{N}{N_{T,j=1}}$ allocations satisfying this constraint:

$$P(\mathbf{W} \mid \mathbf{X}, \mathbf{Y}(0), \mathbf{Y}(1)) = \begin{cases} \binom{N}{N_{T,j=1}}^{-1} & \text{if } \sum_{i=1}^{N} W_{i,j=1} = N_{T,j=1} \text{ and } W_{i,j=2} = 1 - W_{i,j=1}, \\ 0 & \text{otherwise.} \end{cases}$$

**Key structural insight.** Both mechanisms enforce $W_{i,j=2} = 1 - W_{i,j=1}$. This means the assignment in period 2 is entirely determined by the assignment in period 1. The randomness is entirely in who gets treatment *first*.

## 1.4 The Science Table and Causal Estimands

Let $Y_{i,j}(W_{i,j} = w)$ denote participant $i$'s potential outcome at period $j$ when assigned $W_{i,j} = w$. Each participant has four potential outcomes:

| Unit $i$ | $Y_{i,j=1}(0)$ | $Y_{i,j=1}(1)$ | $Y_{i,j=2}(0)$ | $Y_{i,j=2}(1)$ |
|---|---|---|---|---|
| 1 | $Y_{1,j=1}(0)$ | $Y_{1,j=1}(1)$ | $Y_{1,j=2}(0)$ | $Y_{1,j=2}(1)$ |
| 2 | $Y_{2,j=1}(0)$ | $Y_{2,j=1}(1)$ | $Y_{2,j=2}(0)$ | $Y_{2,j=2}(1)$ |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ |
| $N$ | $Y_{N,j=1}(0)$ | $Y_{N,j=1}(1)$ | $Y_{N,j=2}(0)$ | $Y_{N,j=2}(1)$ |

Because of the crossover constraint, for each participant $i$, exactly **two** of the four potential outcomes are observed: one from period 1 and one from period 2, corresponding to the two different treatments they receive. The other two remain counterfactual.

The **observed outcome** is written compactly using the consistency equation:

$$Y_{i,j}^{obs} = W_{i,j}\, Y_{i,j}(1) + (1 - W_{i,j})\, Y_{i,j}(0).$$

**Causal estimands** defined from the science table:

- **Unit-period causal effect:**

  $$\tau_{i,j} = Y_{i,j}(1) - Y_{i,j}(0).$$

  The effect of treatment versus control for participant $i$ in period $j$.

- **Period-averaged participant-level effect:**
$$\tau_i = \frac{1}{2}(\tau_{i,1} + \tau_{i,2}).$$
The average effect of treatment on participant $i$, averaging across both periods.
- **Population-average period-level effect:**
$$\tau_j = \frac{1}{N}\sum_{i=1}^{N} \tau_{i,j}.$$
The average effect across all participants within a specific period $j$.
- **Grand average (period- and population-average) effect:**
$$\tau = \frac{1}{2}(\tau_{j=1} + \tau_{j=2}).$$

**A remarkable feature of the crossover design.** In a standard parallel-arm CRE, the individual causal effect $\tau_i = Y_i(1) - Y_i(0)$ is fundamentally unidentifiable: we can never observe both $Y_i(1)$ and $Y_i(0)$ for the same person. In a crossover design, participant $i$ is observed under *both* treatment and control (in different periods), so $\tau_i$ is estimable directly from data - provided there are no carry-over or time effects (discussed below). This is the deepest advantage of the crossover design over all between-person designs.

## 1.5 The Observed Table

For a participant with $W_i = (0, 1)$ (control first, treatment second):

- Period 1 observed: $Y_{i,j=1}(0)$. Unobserved: $Y_{i,j=1}(1)$.
- Period 2 observed: $Y_{i,j=2}(1)$. Unobserved: $Y_{i,j=2}(0)$.

For a participant with $W_i = (1, 0)$ (treatment first, control second):

- Period 1 observed: $Y_{i,j=1}(1)$. Unobserved: $Y_{i,j=1}(0)$.
- Period 2 observed: $Y_{i,j=2}(0)$. Unobserved: $Y_{i,j=2}(1)$.

In either case, participant $i$ contributes one observed outcome under treatment and one under control, just in different periods.

## 1.6 Fisherian Inference

The sharp null hypothesis states that treatment has no effect on any participant in either period:
$$H_0 : Y_{i,j}(1) = Y_{i,j}(0) \quad \text{for all } i = 1, \ldots, N \text{ and } j = 1, 2.$$

Under $H_0$, all four potential outcomes for each participant equal their observed values, and the entire science table is known.

For each participant $i$, define the **individual difference**:

$$d_i = \begin{cases} Y_{i,j=1}(1) - Y_{i,j=2}(0) & \text{if } W_i = (1,0), \\ Y_{i,j=2}(1) - Y_{i,j=1}(0) & \text{if } W_i = (0,1). \end{cases}$$

In both cases, $d_i$ is (treatment period outcome) $-$ (control period outcome) for participant $i$, regardless of which period the treatment was received.

The **test statistic** is the paired $t$-statistic:

$$T_{\text{paired}} = \frac{\frac{1}{N}\sum_{i=1}^{N} d_i}{s_D/\sqrt{N}}, \qquad s_D^2 = \frac{1}{N-1}\sum_{i=1}^{N}\left(d_i - \frac{1}{N}\sum_{i=1}^{N} d_i\right)^2.$$

**How the reference distribution is computed.** Under the sharp null, all potential outcomes are fixed. The only randomness is in $\mathbf{W}$: specifically, in which participants receive sequence $(1,0)$ versus $(0,1)$. For each of the $2^N$ (or $\binom{N}{N_T}$) valid assignments, one can recompute $T_{\text{paired}}$. The p-value is the fraction of assignments that produce a test statistic as or more extreme than the observed one. No distributional assumption is required.

## 1.7 Neymanian Inference

The natural point estimator for the individual-level effect $\tau_i$ is simply:

$$\hat{\tau}_i = d_i,$$

since $d_i$ is exactly (treatment outcome) $-$ (control outcome) for participant $i$. This is a complete and unbiased estimate of $\tau_i$ under no carry-over and no time effect assumptions.

The natural estimator for the grand average $\tau$ is:

$$\hat{\tau} = \bar{d} = \frac{1}{N}\sum_{i=1}^{N} d_i.$$

## 1.8 Carry-over and Time Effects

The primary threat to validity in crossover designs is the violation of the assumption that the two periods are exchangeable, that the potential outcome at period 2 is not influenced by the treatment received at period 1.

- **Carry-over effect:** The treatment from period 1 continues to affect the participant during period 2. If carry-over is present, the "control" observation in period 2 for a $(1,0)$ participant is actually the outcome under some residual treatment effect, biasing $d_i$.

- **Time effect (period effect):** Outcomes may change over time for reasons unrelated to treatment. Learning, seasonal variation, disease progression. If both sequences experience the same time trend, this cancels in $d_i$; but if it interacts with treatment sequence, it can bias estimates.

**Mitigation strategies.**

- **Washout period:** Insert a sufficiently long gap between periods to allow the treatment effect to dissipate completely. The cannabis driving trial used a one-month washout. The appropriate length depends on the pharmacokinetics of the treatment.
- **Balanced sequences:** Assigning equal numbers to sequences $(1, 0)$ and $(0, 1)$ means that carry-over and time effects act symmetrically on both groups and partially cancel when averaging $d_i$ across participants.

**When crossover designs should not be used.** If the treatment effect is permanent or very long-lasting (e.g., a surgical intervention, a vaccine), no washout period can eliminate carry-over, and a crossover design is inappropriate. The design also cannot be used when the condition being treated is itself transient: if a patient recovers spontaneously between periods, the second period observation is no longer comparable. In these settings, a parallel-arm design is preferred.

# 2 Part II: Block Randomized Experiments

## 2.1 Motivation: Fisher's Rule

In a completely randomized experiment with $N$ units, randomization balances all covariates *in expectation*, but any particular realization of randomization may produce imbalanced groups. If the outcome is strongly related to a known covariate (e.g., sex, age group, clinic site), accidental imbalance inflates variance and can confound inference.

**Blocking** (also called stratification) prevents this by guaranteeing covariate balance *by design*: the researcher partitions units into groups (blocks) of similar units based on a known covariate, and then randomizes treatment *separately within each block*. This is captured in Fisher's famous dictum:

*"Block what you can, randomize what you cannot."*

**Why blocking improves precision.** Suppose males and females differ substantially in their baseline outcomes. In a CRE, between-person variability (including this sex difference) contributes to estimation error. By blocking on sex and randomizing within each sex separately, the sex effect is completely removed from the error term. The variance of $\hat{\tau}$ in a block design is determined only by within-block variation, not between-block variation. The larger the between-block differences, the greater the precision gain from blocking.

**Why blocking is sometimes difficult.** Blocking requires that the blocking variable be measured *before* randomization. If the variable of interest (e.g., a biomarker, a psychological trait) is expensive or time-consuming to measure, pre-randomization blocking may not be feasible. **Rerandomization** is a computationally intensive modern alternative: draw many candidate randomizations and select one that achieves acceptable covariate balance, without needing to define formal blocks in advance.

## 2.2 Taxonomy of Block Designs

- **Complete block design:** Each block is large enough to contain at least one unit per treatment level. With two treatments, each block needs at least two units.
- **Incomplete block design:** Some blocks are smaller than the number of treatment levels, so not every treatment is represented in every block.
- **Balanced design:** The number of units assigned to each treatment is the same within every block.

Balanced complete block designs have attractive mathematical properties, including an orthogonal decomposition of total variability into treatment and block components.

## 2.3 Setup and Notation

For clarity, we work with the simplest non-trivial case from the slides (Section 9.3.1 of the textbook):

- One treatment factor with two levels: treatment ($W_i = 1$) and control ($W_i = 0$).
- One blocking variable with two levels: female ($k = f$) and male ($k = m$).
- $N = N(f) + N(m)$ total units.
- Within block $k$, exactly $N_t(k)$ units are assigned to treatment and $N(k) - N_t(k)$ to control.

## 2.4 The Assignment Mechanism

The assignment mechanism reflects independent completely randomized experiments within each block:

$$P(\mathbf{W} \mid \mathbf{X}, \mathbf{Y}(0), \mathbf{Y}(1)) = \begin{cases} \binom{N(f)}{N_t(f)}^{-1} \binom{N(m)}{N_t(m)}^{-1} & \text{for } \mathbf{W} \in \mathbb{W}^+, \\ 0 & \text{otherwise,} \end{cases}$$

where:

$$\mathbb{W}^+ = \left\{ \mathbf{W} : \sum_{i \in \text{Block } f} W_i = N_t(f) \;\text{ and }\; \sum_{i \in \text{Block } m} W_i = N_t(m) \right\}.$$

The total number of valid assignments is $\binom{N(f)}{N_t(f)} \times \binom{N(m)}{N_t(m)}$, which is the *product* of two independent binomial counts, one per block.

**Structural interpretation.** The block design is simply two independent completely randomized experiments running in parallel: one within females, one within males. The assignment mechanism factorizes as a product, reflecting this independence. The critical implication for inference is that the p-value under Fisherian inference must be computed using only within-block rerandomizations: a male unit can never be reassigned to the female block.

## 2.5 The Science Table and Causal Estimands

The science table for a block design is similar to the CRE science table, but now each row carries a block label:

| Unit $i$ | Block $k$ | $Y_i(W_i = 0)$ | $Y_i(W_i = 1)$ |
|---|---|---|---|
| 1 | $f$ | $Y_1(0)$ | $Y_1(1)$ |
| 2 | $f$ | $Y_2(0)$ | $Y_2(1)$ |
| 3 | $m$ | $Y_3(0)$ | $Y_3(1)$ |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ |
| $N$ | $m$ | $Y_N(0)$ | $Y_N(1)$ |

For each unit $i$, only one of the two potential outcomes is observed (the one corresponding to the treatment actually received). The observed outcome satisfies the consistency equation:

$$Y_i^{obs} = W_i\, Y_i(1) + (1 - W_i)\, Y_i(0).$$

Three key causal estimands, building in scope:

- **Unit-level effect within block $k$:**
$$\tau_i(k) = Y_i(1) - Y_i(0), \quad i \in \text{Block } k.$$
- **Block-average effect within block $k$:**
$$\tau(k) = \frac{1}{N(k)} \sum_{i \in \text{Block } k} \tau_i(k).$$
  This is the average treatment effect among all units belonging to block $k$.
- **Population-weighted overall average effect:**
$$\tau = \frac{N(f)}{N(f) + N(m)}\,\tau(f) + \frac{N(m)}{N(f) + N(m)}\,\tau(m).$$
  This is a block-size-weighted combination of block-level effects. It equals the treatment effect you would observe in the full population.

**Why the weights in $\tau$ are population-size weights, not equal weights.** If block $f$ has 80 females and block $m$ has 20 males, then females represent 80% of the population. The overall effect should reflect this: $\tau = 0.8\,\tau(f) + 0.2\,\tau(m)$. Averaging $\tau(f)$ and $\tau(m)$ with equal weight 1/2 would answer the question "what is the average effect across blocks?" rather than "what is the average effect across people?", a subtly different and usually less interesting question.

## 2.6 Fisherian Inference

The sharp null hypothesis is:

$$H_0 : Y_i(1) = Y_i(0) \quad \text{for all } i = 1, \ldots, N.$$

Under $H_0$, all potential outcomes are observed, and the reference distribution is obtained by rerandomizing $\mathbf{W}$ within the set $\mathbb{W}^+$ (preserving block assignments).

The natural test statistics are the within-block differences in observed means:

$$\overline{Y}_T^{obs}(f) - \overline{Y}_C^{obs}(f), \qquad \overline{Y}_T^{obs}(m) - \overline{Y}_C^{obs}(m).$$

These can be combined into a single test statistic using a weighted average:

$$T_\lambda = \lambda\Big[\overline{Y}_T^{obs}(f) - \overline{Y}_C^{obs}(f)\Big] + (1-\lambda)\Big[\overline{Y}_T^{obs}(m) - \overline{Y}_C^{obs}(m)\Big], \quad \lambda \in [0,1].$$

The natural choice is $\lambda = N(f)/N$, which corresponds to testing the null against the population-weighted alternative and mirrors the target estimand $\tau$.

## 2.7 Neymanian Inference

### 2.7.1 Estimating the Block-Level ATE $\tau(k)$

The natural estimator is the difference in observed means within block $k$:

$$\hat{\tau}(k) = \overline{Y}_T^{obs}(k) - \overline{Y}_C^{obs}(k).$$

This is unbiased for $\tau(k)$: $\mathbb{E}[\hat{\tau}(k)] = \tau(k)$.

The **exact sampling variance** of $\hat{\tau}(k)$ is:

$$\text{Var}\big(\hat{\tau}(k)\big) = \frac{S_T^2(k)}{N_T(k)} + \frac{S_C^2(k)}{N_C(k)} - \frac{S_{TC}^2(k)}{N(k)},$$

where $S_T^2(k)$ and $S_C^2(k)$ are the population variances of potential outcomes under treatment and control within block $k$, and $S_{TC}^2(k)$ is the variance of individual treatment effects $\tau_i(k)$ within block $k$. As always, $S_{TC}^2(k)$ is unobservable.

The **conservative estimated variance** (assuming $S_{TC}^2(k) = 0$) is:

$$\widehat{\text{Var}}\big(\hat{\tau}(k)\big) = \frac{s_T^2(k)}{N_T(k)} + \frac{s_C^2(k)}{N_C(k)},$$

where $s_T^2(k)$ and $s_C^2(k)$ are the observed sample variances within the treatment and control arms of block $k$, computed as:

$$s_T^2(k) = \frac{1}{N_T(k)-1} \sum_{\substack{i\in\text{Block } k\\ W_i=1}} \left(Y_i^{obs} - \overline{Y}_T^{obs}(k)\right)^2, \qquad s_C^2(k) = \frac{1}{N_C(k)-1} \sum_{\substack{i\in\text{Block } k\\ W_i=0}} \left(Y_i^{obs} - \overline{Y}_C^{obs}(k)\right)^2.$$

In words: $s_T^2(k)$ is the standard sample variance of observed outcomes among *treated* units in block $k$, and $s_C^2(k)$ is the same quantity among *control* units in block $k$.

### 2.7.2 Estimating the Overall ATE $\tau$

The block-weighted estimator of the overall effect is:

$$\hat{\tau} = \frac{N(f)}{N}\left[\overline{Y}_T^{obs}(f) - \overline{Y}_C^{obs}(f)\right] + \frac{N(m)}{N}\left[\overline{Y}_T^{obs}(m) - \overline{Y}_C^{obs}(m)\right].$$

This is unbiased for $\tau$.

The **exact sampling variance** of $\hat{\tau}$ is:

$$\begin{aligned}\text{Var}(\hat{\tau}) = &\left(\frac{N(f)}{N}\right)^2\left[\frac{S_T^2(f)}{N_T(f)} + \frac{S_C^2(f)}{N_C(f)} - \frac{S_{TC}^2(f)}{N(f)}\right] \\ &+ \left(\frac{N(m)}{N}\right)^2\left[\frac{S_T^2(m)}{N_T(m)} + \frac{S_C^2(m)}{N_C(m)} - \frac{S_{TC}^2(m)}{N(m)}\right].\end{aligned}$$

The **conservative estimated variance** (assuming $S_{TC}^2(k) = 0$ for all $k$) is:

$$\widehat{\text{Var}}(\hat{\tau}) = \left(\frac{N(f)}{N}\right)^2\left[\frac{s_T^2(f)}{N_T(f)} + \frac{s_C^2(f)}{N_C(f)}\right] + \left(\frac{N(m)}{N}\right)^2\left[\frac{s_T^2(m)}{N_T(m)} + \frac{s_C^2(m)}{N_C(m)}\right].$$

**The efficiency gain from blocking, made precise.** The total variance $\text{Var}(\hat{\tau})$ is a weighted sum of *within-block* variances only. The between-block variance, the part driven by males and females having different average outcomes, contributes nothing to the sampling error of $\hat{\tau}$, because block membership is known with certainty to the analyst. This is why blocking improves precision when blocks differ substantially in their baseline outcomes: it eliminates a source of variability from estimation entirely.

## 2.8 Effect Modification: Estimating $\tau(m) - \tau(f)$

Beyond estimating an overall average effect, one may ask: does the treatment work *differently* for males and females? This is the question of **effect modification** (or treatment heterogeneity by block).

The estimand is $\tau(m) - \tau(f)$, and the natural unbiased estimator is:

$$\hat{\tau}(m) - \hat{\tau}(f) = \left[\overline{Y}_T^{obs}(f) - \overline{Y}_C^{obs}(f)\right] - \left[\overline{Y}_T^{obs}(m) - \overline{Y}_C^{obs}(m)\right].$$

Since the two blocks are randomized independently, the conservative estimated variance is simply the sum of the within-block estimated variances:

$$\widehat{\mathrm{Var}}\big(\hat{\tau}(m) - \hat{\tau}(f)\big) = \frac{s_T^2(f)}{N_T(f)} + \frac{s_C^2(f)}{N_C(f)} + \frac{s_T^2(m)}{N_T(m)} + \frac{s_C^2(m)}{N_C(m)}.$$

**Why effect modification matters.** Suppose a drug reduces blood pressure by an average of 5 mmHg in the overall population. The block design reveals that it reduces it by 10 mmHg in females but by 0 mmHg in males. The overall average $\tau = 5$ is not misleading if the goal is to understand population-level policy impact, but it conceals the fact that the drug is beneficial only for females. Detecting and reporting $\tau(m) - \tau(f)$ is essential for personalizing treatment recommendations and for understanding the biological mechanisms of the intervention.

# 3 Comparison of Designs

| | **Matched-Pair** | **Crossover** | **Block** |
|---|---|---|---|
| **Unit of analysis** | Pair | Participant | Individual |
| **How variability is reduced** | Pair similar individuals; compare within pair | Each person observed under both treatments | Randomize within pre-defined strata |
| **Individual effect identifiable?** | No (still only one observation per arm per pair) | Yes (observed under both conditions) | No |
| **Main validity threat** | Matching quality; remaining confounding | Carry-over and time effects | Block variable must be measured pre-randomization |
| **Common application** | Observational studies; rare lab settings | Pain, symptom, pharmacokinetic trials | Trials with known demographic or site heterogeneity |

# 4 Practice Problems

See the .R file attached.